



POLITECNICO DI TORINO

MSC MECHATRONICS ENGINEERING

NETWORKED CONTROL SYSTEM

CHARACTERISTICS OF SUCCESSFUL OPINION LEADERS
IN A BOUNDED CONFIDENCE MODEL

PROF:

ANTON PROSKURNIKOV

GROUP:

PIERLUIGI SACCONI, 355355

GOFFREDO PIAZZA, 360031

MICHELE GAZZILLI, 360623

CESARE MESSA, 361616

PERCENTAGE OF WORK:

PIERLUIGI SACCONI, 25%

GOFFREDO PIAZZA, 25%

MICHELE GAZZILLI, 25%

CESARE MESSA, 25%

1 Introduction

The study of opinion dynamics focuses on how individuals in a social network update their beliefs through interaction with peers. Among the various mathematical frameworks proposed, the **Hegselmann-Krause (HK) model** is one of the most prominent agent-based models for continuous opinion formation. It relies on the concept of *bounded confidence*: an agent only considers the opinions of others if they differ by less than a certain threshold T .

While classical studies often focus on consensus reaching in homogeneous groups, real-world scenarios are characterized by the presence of influential actors, or *opinion leaders*, who possess the power to shape public debate. This project builds specifically upon the work of **Chen et al. (2016)**, which investigates the impact of competing opinion leaders on a social group. Chen et al. introduced a parameterized framework defining four key characteristics of leadership: *Reputation*, *Stubbornness*, *Appeal*, and *Extremeness*.

The aim of this work is to validate the findings of Chen et al. and extend their analysis to more complex, asymmetric scenarios.

2 The HK Model with Opinion Leaders

We adopt the extended Hegselmann-Krause model formulation as described in Chen et al. (2016). Consider a society of agents $n = 200$, with two competing opinion leaders L_1 and L_2 . The initial opinion of agent i at time t is represented by a random real value $x_i(t) \in [0, 1]$, while for the two Leaders the initial opinion is fixed at 0.2 and 0.8, respectively for the leader L_1 and L_2 . In the classic HK model, at each time step t , agent i updates its opinion by averaging the opinions of all agents within its *confidence set* $I(i, t)$, defined as:

$$I(i, t) = \{j : |x_i(t) - x_j(t)| \leq T\} \quad (1)$$

where T is the bound of confidence in opinion.

To account for heterogeneity, specific weights (Reputation) are assigned to agents. The update rule is given by the weighted average:

$$x_i(t+1) = \frac{\sum_{j \in I(i, t)} R_j x_j(t)}{\sum_{j \in I(i, t)} R_j} \quad (2)$$

where R_j represents the reputation of agent $j \in I(i, t)$.

2.1 Characteristics of Opinion Leaders

1. **Reputation (R):** Leaders are assigned a weight $R_i \gg 1$, whereas normal agents have a weight $R = 1$.
2. **Stubbornness (S):** This measures the leader's resistance on being influenced. It is related to the leader's confidence bound T_L by the relation:

$$S_i = 1 - T_{L_i} \quad (3)$$

A completely stubborn leader ($S = 1, T_L \approx 0$) never changes their opinion, regardless of the followers' positions. A flexible leader ($S < 1$) may adapt towards the group mean.

3. **Appeal (A):** While T is the confidence range of normal agents towards each other, A_i is the specific confidence range of normal agents towards leader L_i . A high appeal ($A_i > T$) implies high visibility, allowing the leader to influence distant agents who would otherwise ignore peers at that distance.
4. **Extremeness (E):** This quantifies how polarized a leader is relative to the neutral center (0.5). It is defined as:

$$E_i = 2 \cdot |0.5 - x_{L_i}(t)| \quad (4)$$

where $E = 0$ indicates a centrist leader and $E = 1$ indicates a leader at the boundary (0 or 1).

2.2 Impact of Group bound of confidence

For each variable, we study its behavior at different values of the group bound of confidence T . This is because much higher T is, much faster consensus will be reached. To demonstrate this, we made some simulations for T that goes from 0.1 to 0.5 in step of 0.2, while all the other variables are fixed at 0.5, and both reputations of the leaders $R1$ and $R2$ are settled at 10. Indeed, we can see in Figure 1 that the time instant when consensus is reached the smaller it is, the bigger T is.

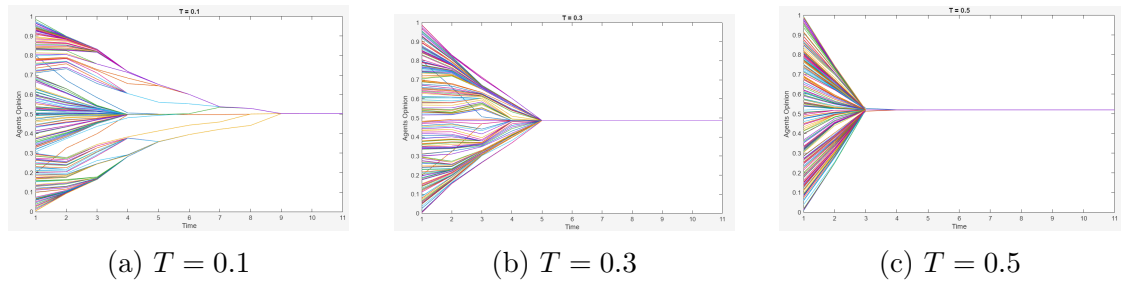


Figure 1: Configuration of final opinion at different values of T .

3 Validation of Reputation impact

To analyze the impact of leader's reputation, while the one of all the other agents is set at 1, we let $R1$ and $R2$ variate from 1 to 400 in steps of 10, while the group bound of confidence T varying from 0.05 to 0.5 in steps of 0.01, and we set $T_{L_i} = A_i = T$, and $E = 0.6$ for both leaders .

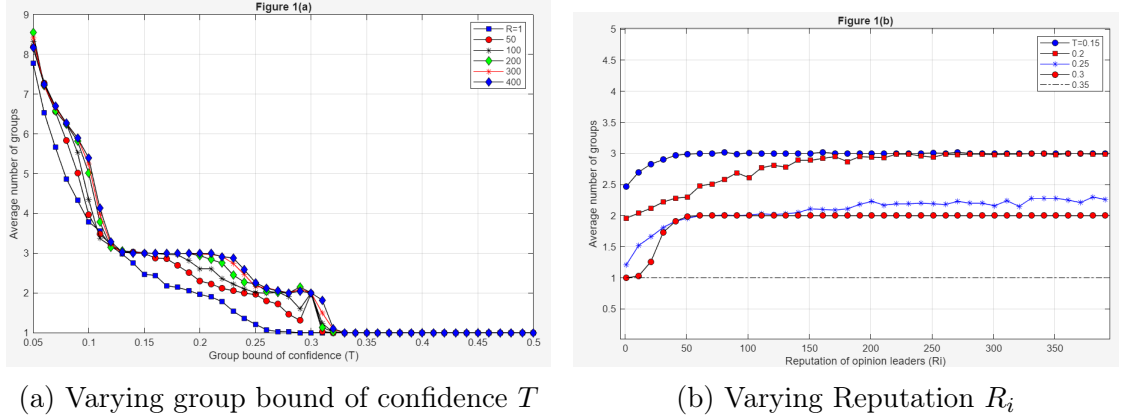


Figure 2: Average number of groups with respect to: (a) T for different R , and (b) R_i with different T

We can see on Fig.1(a) that, for $T \in (0.05, 0.1)$, we have the formation of more than 3 groups, and thus we are far from consensus. For $T \in (0.1, 0.3)$ we have from 3 groups down to 1 group formed, but just for $R_i = 1$. Consensus has not been reached yet. For $T \in (0.32, 0.5)$, only 1 group is formed for each value of R_i , and so consensus is completely reached. This happens because between the epistemic range of the two leaders there will be an overlap, and thus the two leaders start to interact with each other. To better highlight this, we can have a look at Figure 2 where we plotted the average number of followers of the two opinion leaders, by fixing the reputation of opinion leader $R2 = 50$ and varying the reputation of leader $L1$ from 1 to 400 in steps of 5, while keeping the values of other characteristics the same as in the previous simulations, and we variate $T \in (0.2, 0.4)$. Our results are comparable with the one on the paper.

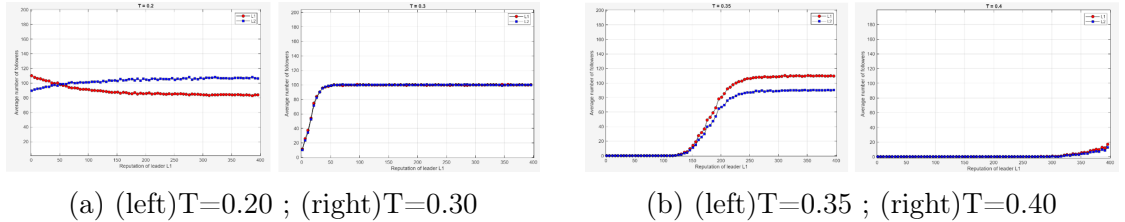


Figure 3: Average number of followers of opinion leaders $L1$ and $L2$ with respect to the reputation of leader $L1$, where $R2 = 50$.

4 Validation of Stubbornness Impact

To assess the reliability of the simulation results, the numerical implementation was validated by reproducing the plots reported in Section 3.2 of the reference paper. The obtained results show qualitative agreement with the original ones, confirming the correctness of the implemented model.

Minor discrepancies are mainly due to the reduced simulation horizon adopted to limit computational cost and do not affect the overall trends. Therefore, the qualitative consistency with the reference plots is sufficient to ensure the reliability of the simulation framework and of the results presented in the following sections.

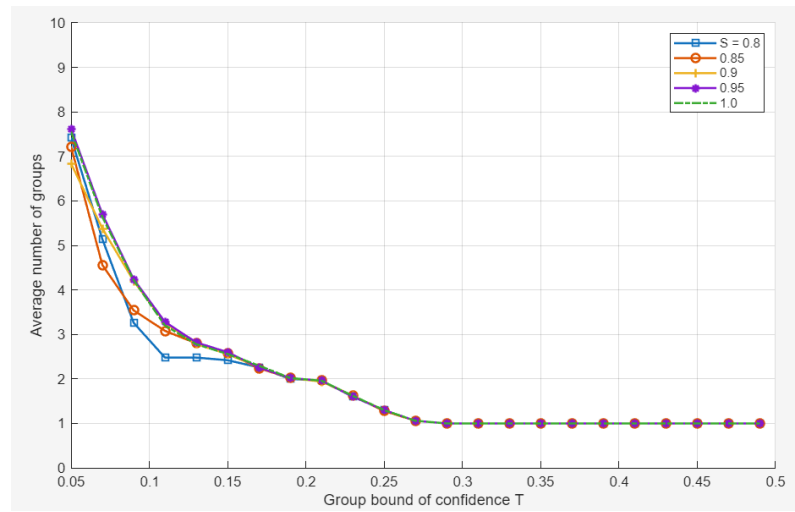


Figure 4: Average number of groups with respect to group bound of confidence T with different stubbornness values, and stubbornness S_i with different bounds of confidence, where $R_i = 1$, $A_i = T$, $E_i = 0.6$. Averages taken over 50 runs

The plot shows that the stubbornness values do not have an huge impact on the average number of groups, in fact all the curves corresponding to the different values of stubbornness of the two leaders have approximately the same trend.

4.1 Analysis of transient behavior

In the reference paper, with regard to stubbornness, the analysis is carried out considering the system at steady state, and the results are discussed as a function of the stubbornness parameter. In this portion of the report, instead, the transient behavior of the system is analyzed, with a specific focus on the time required for the system to reach the steady state, that is, the condition in which stable opinion groups are formed and no further variations occur.

In order to extract this information, the convergence time was defined as the first iteration at which the maximum variation of agents' opinions between two consecutive time steps falls below a fixed threshold. In particular, the convergence

time is computed as:

$$t_{\text{conv}} = \min \left\{ t : \max_i |x_i(t+1) - x_i(t)| < \varepsilon \right\} \quad (5)$$

where ε is a small positive tolerance.

The reported plot represents the scenario in which the stubbornness of leader L_2 is fixed to 0.95, while the stubbornness of leader L_1 is varied from 0.8 to 1 with a step of 0.01. The graph allows one to visualize the number of iterations required for the system to reach stability.

The plot can be divided into three distinct regions. For values of S_1 between 0.8 and 0.9, the number of required iterations remains approximately constant. In the interval between 0.9 and 0.95, the convergence time slightly increases as the stubbornness increases. Finally, starting from $S_1 = 0.95$, which corresponds to the condition in which both leaders have the same stubbornness value, the number of iterations required for convergence grows rapidly. It is worth noting that the convergence time increases from approximately 24 iterations for $S_1 = 0.8$ up to about 463 iterations for $S_1 = 1$.

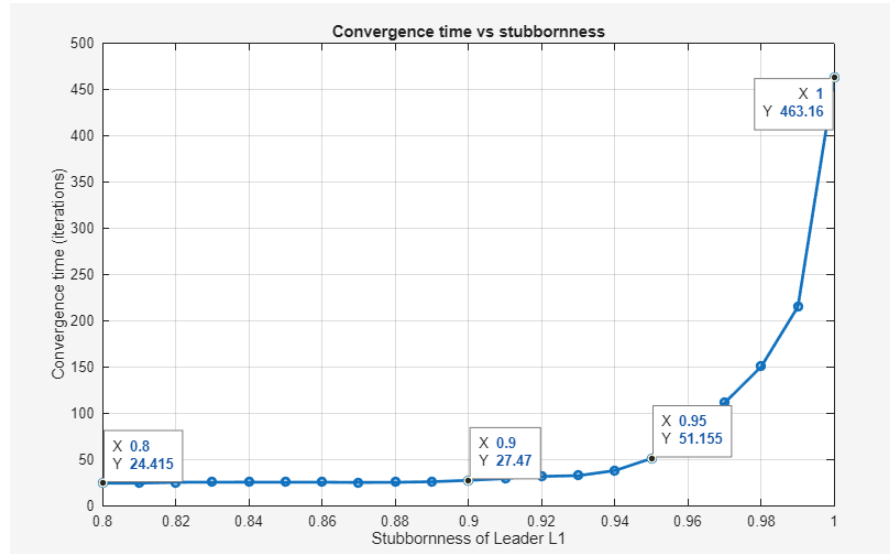


Figure 5: Number of iterations required to reach the steady state as the stubbornness S_1 of leader L_1 varies in the range $[0.8, 1.0]$ with a step of 0.01.

5 Validation of Appeal Impact

In this paragraph it has been highlighted the impact of leaders' *Appeal* (A_i) on group opinions dynamics and the formation of group of followers. The replicated model correctly captures all significant phenomenological behaviors and the correspondence confirms the robustness of the underlying algorithm.

The first plot regarded the replication of the results obtained in a context with $n = 200$ agents, keeping a fixed *Extremeness* (E_i) = 0.6 with leaders' opinions equals to $L_1 = 0.2$ and $L_2 = 0.8$, varying simultaneously both the leaders from 0.25 to 0.5 in steps of 0.01 and the group bound of confidence T from 0.05 to 0.5 in steps of 0.01, while $R_i = 50$ and $S_i = 1$ ($i = 1, 2$). The following results, shown in the *Figure 6*, fully demonstrate the consistence with the replicated plot underlying the reliability of the simulation environment.

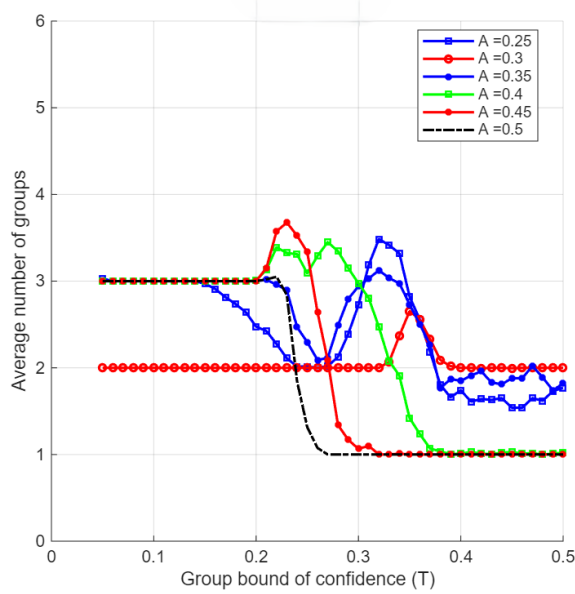


Figure 6: Average number of groups with respect to (group bound of confidence T with different appeal values with different group bounds of confidence with averages taken over 200 runs

Moreover, the second replication focuses the attention on the role of *Appeal* (A_i) on attracting followers by fixing the appeal of opinion leader L_2 to be $A_2 = 0.4$, and varying that of leader L_1 from 0.2 to 0.5 in steps of 0.05, while $R_i = 50$ and $S_i = 1$ ($i = 1, 2$). The results shown in the *Figure 7*, clearly demonstrate the competitive dynamic analyzed in the paper.

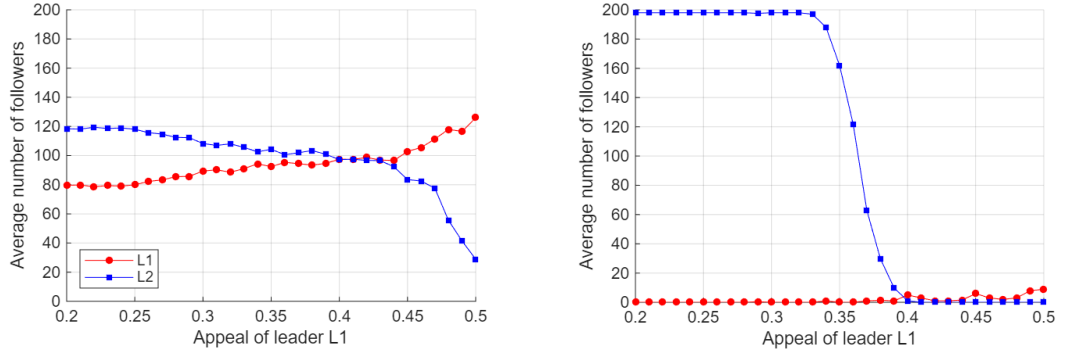


Figure 7: Average number of followers of opinion leaders with respect to $T=0.25$ (left) and $T=0.4$ (right)

5.1 Analysis of Stubbornness' effects on Appeal

In order to be consistent with realistic social dynamics leaders often adjust their positions to align with public opinion. To simulate this, the "stubbornness" constraint was relaxed for Leader 1 ($T_{L1} = 0.2$) while keeping the $T_{L2} = 0$, creating a scenario of a Flexible "Democratic" (L_1) versus a Rigid Leader (L_2).

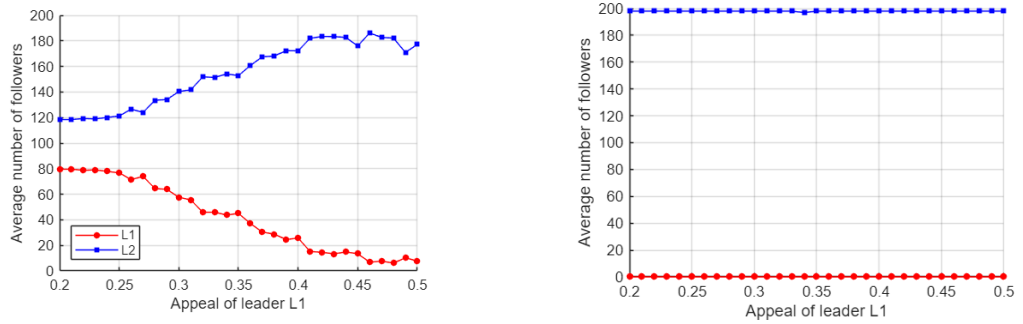


Figure 8: Average group of followers in the presence of a Flexible "Democratic" Leader (L_1) versus a Rigid Leader (L_2) with respect to $T=0.25$ (left) and $T=0.4$ (right)

This experiment demonstrates that stubbornness is a dominant strategy, when reputation is symmetric, to attract the largest number of followers.

6 Validation of Extremeness Impact

To ensure the reliability of our simulation framework, we first replicated the experiments conducted by Chen et al. regarding the impact of *Extremeness* (E) on opinion dynamics. We simulated a system of $n = 200$ agents with two competing leaders, L_1 and L_2 , positioned symmetrically with respect to the center of the opinion space. The leaders' extremeness was varied from 0 to 1, while keeping other characteristics constant ($R = 50$, $S = 1$, $A = 0.4$). Averages taken over 200 runs. Our results fully confirm the findings reported in the reference literature:

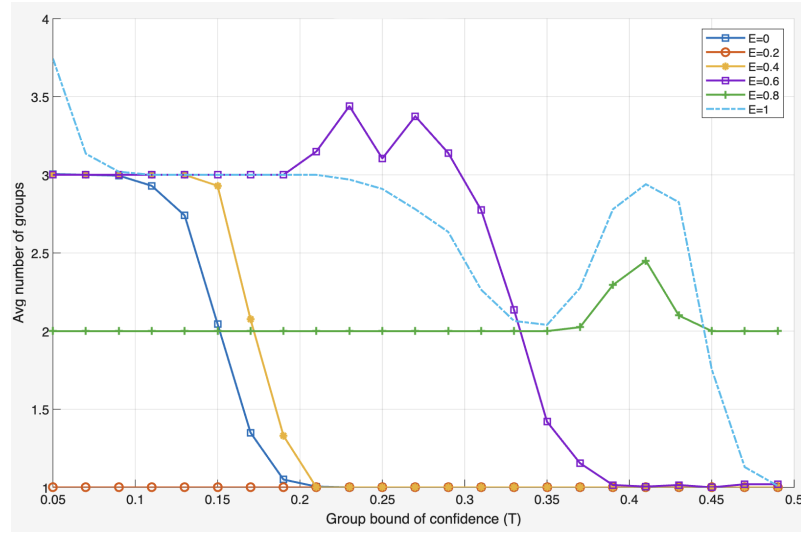


Figure 9: Impact of leader extremeness on group fragmentation. The results confirm that higher extremeness hinders consensus, leading to polarization.

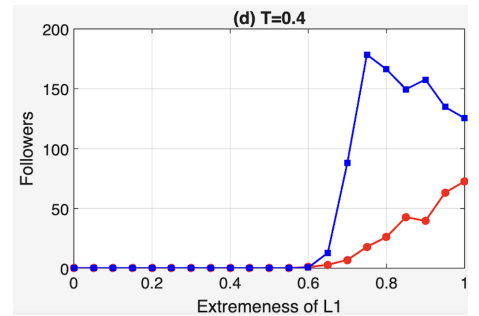
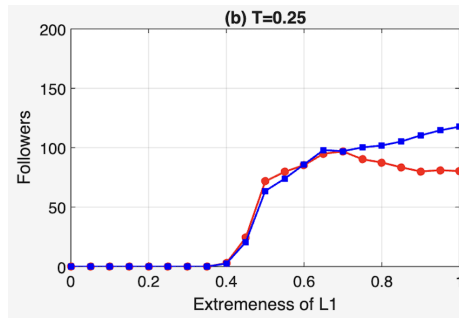
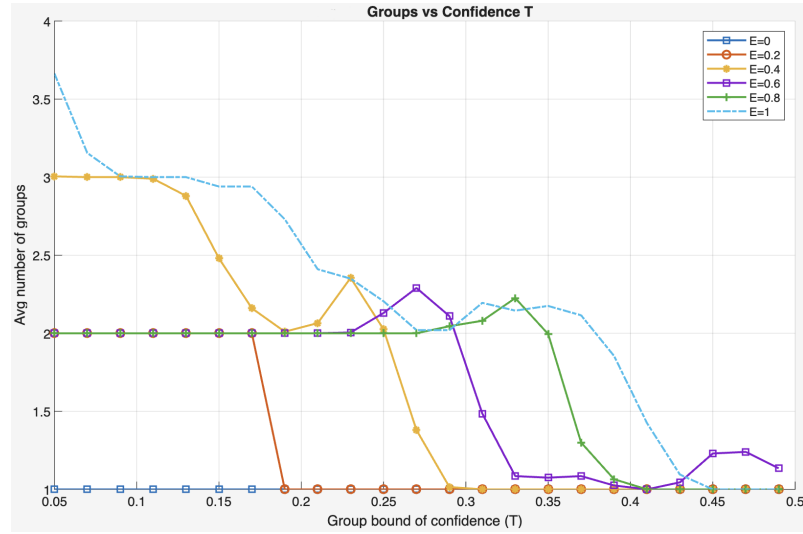


Figure 10: Average number of followers for Leader 1 and Leader 2 as a function of E_1 , with E_2 fixed at 0.6. It is shown that there are some critical extremeness values so that neither leader can attract any follower, but we can see an increasing number of followers of both leaders if E_1 becomes larger than 0.6.

6.1 Asymmetric Leadership

In this experiment, we simulate a scenario often observed in political dynamics:

- **Leader 1 ("The Populist"):** Low Reputation ($R_1 = 10$) but High Appeal ($A_1 = 0.6$).
- **Leader 2 ("The Institution"):** High Reputation ($R_2 = 100$) but Low Appeal ($A_2 = 0.2$).



(a) Avg groups vs Confidence T

Figure 11: Fragmentation analysis in the Asymmetric Scenario. The presence of a High-Appeal leader (L_1) extends the range of stability, maintaining consensus (1 group) for higher values of Extremeness compared to symmetric scenarios.

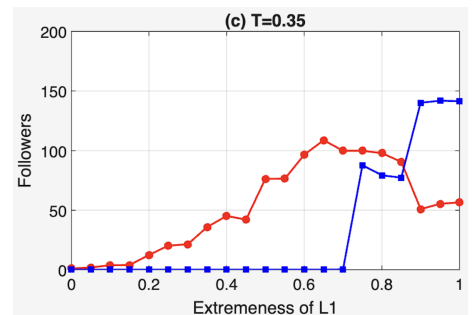
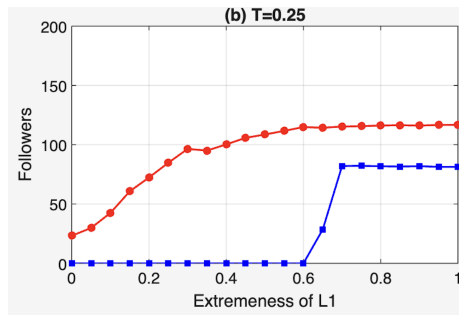


Figure 12: Follower counts in the Asymmetric Scenario. For $T \leq 0.25$, the high-appeal leader (Red) wins. For $T \geq 0.35$, the high-reputation leader (Blue) dominates, provided that extremeness is high enough to trigger polarization.

7 Conclusion and Future Developments

In this report, we successfully validated the extended Hegselmann-Krause model proposed by Chen et al., confirming the critical role of *Reputation*, *Stubbornness*, *Appeal*, and *Extremeness* in shaping opinion dynamics. Our simulations corroborated that while higher Reputation and Appeal generally facilitate follower attraction, high Extremeness acts as a barrier to consensus, triggering polarization.

Beyond the baseline validation, our extended analysis provided three novel insights into the trade-offs inherent in leadership strategies:

- **The Cost of Rigidity (Transient Analysis):** We demonstrated a non-linear relationship between stubbornness and convergence time. As leaders become rigid ($S \rightarrow 1$), the time required to reach a steady state increases exponentially. This highlights a critical efficiency trade-off: a slightly flexible leader allows the group to achieve consensus significantly faster than a completely stubborn one.
- **The Paradox of Adaptability:** Finally, we observed that while flexible leaders can better retain followers in the short term, they risk a "dilution of leadership." By adapting too much, the leader effectively merges with the crowd, losing the ability to steer the consensus direction, whereas a rigid leader exerts a constant, unidirectional pull.
- **Visibility vs. Power (Asymmetric Leadership):** By breaking the symmetry, we identified two distinct connectivity regimes. In fragmented societies ($T \leq 0.25$), a High-Appeal leader dominates. Conversely, in connected networks ($T \geq 0.35$), the structural power of Reputation prevails once polarization is triggered.

7.1 Future Developments

While this report focused on static parameter configurations, several avenues for future research emerge from our findings:

1. **Strategic Positioning:** In our current model, leaders' initial opinions are fixed parameters. A significant extension would be to treat the initial position as a strategic choice: leaders could optimize their starting opinion based on the initial distribution of the population to maximize their follower count.
2. **Dynamic Strategies:** Implementing leaders with time-varying characteristics (e.g., a leader who starts flexible to gain followers and becomes stubborn later to guide them) could model realistic political campaigns more accurately.
3. **External Influence:** Incorporating a "Mass Media" agent as a global, non-interactive field (as suggested by Chen et al.) could simulate the background noise of public information affecting the competition between leaders.